

UEC610: COMPUTER ARCHITECTURE				
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Course Objective: To introduce the concept of parallelism followed in the modern RISC based computers by introducing the basic RISC based DLX architecture. To make the students understand and implement various performance enhancement methods like memory optimization, Multiprocessor configurations, Pipelining and interfacing of I/O structures using interrupts and to enhance the student's ability to evaluate performance of these machines by using evaluation methods like CPU time Equation, MIPS rating and Amdahl's law.				
Syllabus Fundamentals of Modern Computer Design: Historical Evolution of Microprocessors, Basic Functional Units of a Computer and its operational Concepts. Computer Types - Von-Neuman & Harvard Architecture, CISC and RISC architectures. Performance metrics- CPU Time Equation, MIPS Rating, Amdahl's Law and Measure of system Availability (MTTF, MTTR), Control Unit- Hardwired and micro-programmed Control unit. Instruction Set Principles: Classification of Instruction set architectures- Stack, Register and Accumulator based, Addressing modes of DLX architecture, Instruction format , DLX ISA, Coding Examples Effectiveness of DLX. Pipelining and Parallelism: Idea of pipelining, The basic pipeline for DLX processor, the MIPS pipeline Design issues of Pipeline Implementation, Pipeline Hazards-Structural Hazards, Data hazards, Control Hazards, Techniques to remove—Structural Hazards, Overcoming data hazards with Forwarding/ Bypassing and dynamic scheduling, overcoming Control Hazards –Static Techniques – Freeze the pipeline, Prediction Techniques , Advanced DLX hardware Dynamic techniques – 1/2/n bit predictors , Branch delay Slot , Pipeline Scheduling and Loop Unrolling and Scheduling. Multicycle operations, Superscalar DLX Architecture, The VLIW Approach. Memory Hierarchy Design: Introduction to Multilevel Memories, Principle of Temporal and Spatial Locality, Cache memory, Cache Addressing, Cache Organization, Write Policies, Reducing Cache Misses, Cache Associatively Techniques, Reducing Cache Miss Penalty, Reducing Hit Time, Main Memory Technology, Virtual memory and its addressing, Fast Address Translation, Page tables, Multilevel Page table, Translation Lookaside buffer, Crosscutting issues in the design of Memory Hierarchies. Multiprocessors: Characteristics of Multiprocessor Architectures, Flynn's multiprocessor architecture, Multiple Centralized Shared Memory Architectures- Benes and Butterfly network, Distributed Shared Memory Architectures, Cache Coherence, Synchronization, Models of Memory Consistency-Snoopy Protocols. Input/ Output Organization and Buses: Accessing I/O Devices, Interrupts, Handling Multiple Devices, Controlling device Requests, Exceptions, Direct Memory Access, Bus arbitration policies, Synchronous and Asynchronous buses, Parallel port, Serial port, Standard I/O interfaces, Peripheral Component Interconnect (PCI) bus and its architecture, SCSI Bus, Universal Synchronous Bus (USB) Interface. Familiarization with standards: IEEE P802.11, ANSI/IEEE standard 754.				

Laboratory Work:**Course Learning Objectives (CLO)**

After the completion of the course, the students will be able to:

1. Explain the major differences between a RISC and a CISC based processor.
2. Evaluate the performance of a processor with an enhancement applied and analyze it for future application.
3. Identify the concept of parallelism and pipelining and hazard removal in context to the same.
4. Investigate the Memory Hierarchy design, Memory Addressing and Address translation ,
5. Identify Multiprocessor Configuration, Memory Interconnections- Coherence and Input/output devices interfacing to a processor.

Text Books

1. Hennessy, J. L., and D. A. Patterson. *Computer Architecture: A Quantitative Approach*, 2nd ed. San Mateo, CA: Morgan Kaufman, 1995. ISBN: 1558603727
2. Hennessy, J. L., Patterson, D. A., *Computer Architecture: A Quantitative Approach*, Elsevier (2009) 4th ed.
3. Hamacher, V., Carl, Vranesic, Z.G. and Zaky, S.G., *Computer Organization*, McGraw-Hill (2002) 2nd ed.

Reference Books

1. Murdocca, M. J. and Heuring, V.P., *Principles of Computer Architecture*, Prentice Hall (1999) 3rd ed.
2. Stephen, A.S., Halstead, R. H., *Computation Structure*, MIT Press (1999) 2nd ed.